TCX2005: Information Systems, Management and Organisations

L9: Enhancing Decision Making

Prakash C. Sukhwal (Dr.)
Senior Lecturer, DISA
pcsdisa@nus.edu.sg

Lesson Objectives

- Understand different types of decisions and decision-making processes
- Explore business intelligence and analytics capabilities that transform data into actionable insights
- Examine how decision support systems serve different organizational needs

Learning Resources

- Textbook (read before class)
  - Read chapter 12
- Any Other Material
  - Shared on Canvas for this week

Case Study

Read the case study from textbook on –Eastspring: Targeted Enterprise System Building:

Identify:

a. The problem
b. Key Reasons or Causes
c. Solution and its impact
d. Learnings

Note: be ready to share your thoughts

Case Study

Problem:

- End-of-life technology and legacy systems becoming unmanageable
  - Complex environment with 30 key applications in Singapore HQ plus 20+ other applications
- Upgrade projects stalled in 2014 due to system complexity
- Growing systemic problems that appeared intractable
- Existing infrastructure couldn't sustain expansion plans
  - Systems unable to support company's growth demands

Key Reasons:

- In-house developers integrated modified systems to existing legacy systems
- Applications built around the systems over the years to cater to Eastspring's business needs
- Locally specific business needs across regional business units
- Complex applications built around systems to meet Eastspring's evolving business needs

Case Study

Solution: A strategic Approach

- In 2015, all business processes were reviewed
- Created a Target Operating Model (TOM) for the whole region
- Developed a 3-year project plan
    - Strategic approach based on classification:
    - Critical processes ➔ Managed by an in-house team
    - Important processes ➔ Outsourced
    - Unimportant processes ➔ Use off-the-shelf products
- Strategic decision to acquire systems with breadth of services rather than just best-in-breed
- Decision to use off-the-shelf products for all processes, avoiding in-house development
- Required Eastspring to change its business processes to match the new enterprise software

Case Study

Solution Impact

- Expansion to 14 markets worldwide (including UK, US, and Luxembourg)
- 25% growth in funds under management since 2013
- Enhanced risk parameter understanding for various funds
- More sophisticated analysis capabilities replacing basic Excel spreadsheets
- Improved operational efficiency and decision-making capabilities

Learnings

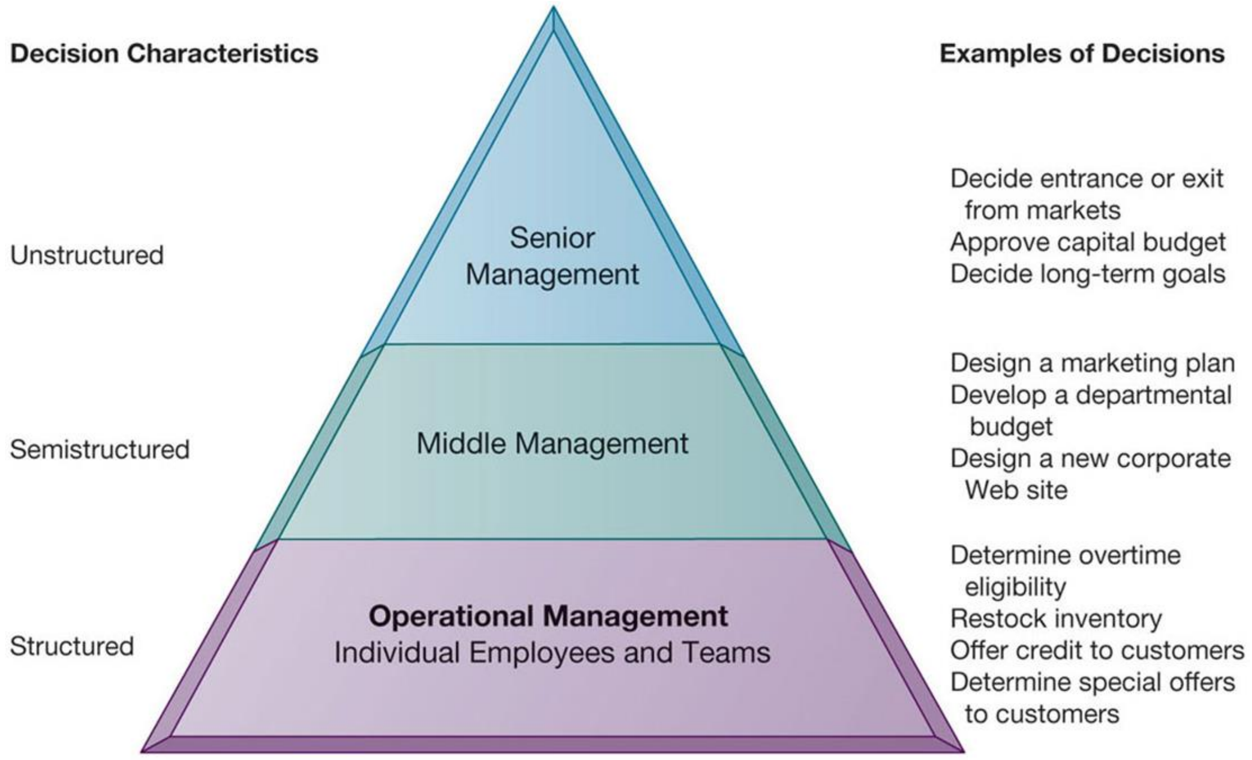
- Business process evaluation should precede technology implementation
    - Categorizing processes by strategic importance enables better resource allocation
- Adapting business processes to standard software can be more effective than customization
- Cloud-based solutions can reduce infrastructure needs and specialized team requirements
- Strategic vendor selection aligned with specific business needs yields better results
- Phased implementation enables learning and adjustment throughout the project

Types of Decisions

- Structured decisions: Repetitive, routine decisions with defined procedures
  - Example: Implementing standard off-the-shelf solutions for unimportant processes like HR functions (in the case study)
- Semi-structured decisions: Some structured elements but requiring human judgment
  - Example: Categorizing business processes as critical, important, or unimportant and determining appropriate management approaches (in the case study)
- Unstructured decisions: No predetermined procedures, relying heavily on intuition and experience
  - Example: Strategic decision to completely overhaul technology and adapt business processes to match new enterprise software (in the case study)

Types of Decisions (cont.)

- Senior managers
  - Make many unstructured decisions
- Middle managers
  - Make more structured decisions but these may include unstructured components
- Operational managers and rank and file employees
  - Make more structured decisions



Activity

For an airlines company; each decision scenario below, identify:

- Who would likely make this decision (Senior Management, Middle Management, or Operational Management)
- What type of decision it is (Structured, Semi-structured, or Unstructured)

#	Decision Scenario	Who Makes It?	Decision Type
1	Whether to expand airline routes to South America		
2	Determining daily crew assignments for scheduled flights		
3	Selecting the appropriate aircraft type for a particular route		
4	Setting ticket pricing strategy for the next fiscal year		
5	Approving an emergency flight diversion due to weather		
6	Deciding whether to merge with a regional competitor		
7	Scheduling routine aircraft maintenance		
8	Optimizing connecting flight timing at hub airports		
9	Determining whether to invest in next-generation aircraft		
10	Approving overtime for ground staff during peak season		

Activity

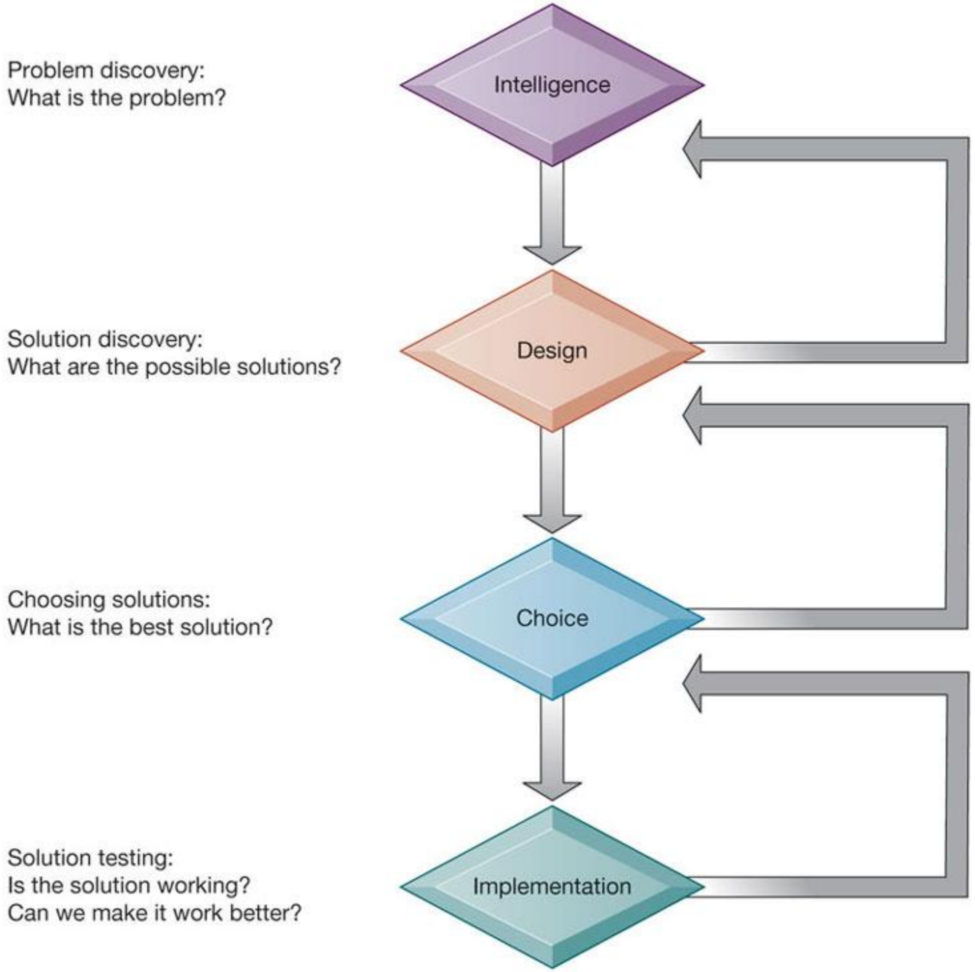
For an airlines company; each decision scenario below, identify:

- Who would likely make this decision (Senior Management, Middle Management, or Operational Management)
- What type of decision it is (Structured, Semi-structured, or Unstructured)

#	Decision Scenario	Who Makes It?	Decision Type
1	Whether to expand airline routes to South America	Senior Management	Unstructured
2	Determining daily crew assignments for scheduled flights	Operational Management	Structured
3	Selecting the appropriate aircraft type for a particular route	Middle Management	Semi-structured
4	Setting ticket pricing strategy for the next fiscal year	Senior Management	Semi-structured
5	Approving an emergency flight diversion due to weather	Operational Management	Semi-structured
6	Deciding whether to merge with a regional competitor	Senior Management	Unstructured
7	Scheduling routine aircraft maintenance	Operational Management	Structured
8	Optimizing connecting flight timing at hub airports	Middle Management	Semi-structured
9	Determining whether to invest in next-generation aircraft	Senior Management	Unstructured
10	Approving overtime for ground staff during peak season	Middle Management	Structured

The Decision-Making Process

- Intelligence: Discovering, identifying, and understanding the problems occurring in the organization
- Design: Identifying and exploring solutions to the problem
- Choice: Choosing among solution alternatives
- Implementation: Making chosen alternative work and continuing to monitor how well solution is working



Real-World Decision Making

Three main reasons why investments in IT/IS do not always produce positive results

- Information quality
  - High-quality decisions require high-quality information
- Management filters
  - Managers have selective attention and have variety of biases that reject information that does not conform to prior conceptions
- Organizational inertia and politics
  - Strong forces within organizations resist making decisions calling for major change

High-Velocity Automated Decision Making

- Made possible through computer algorithms precisely defining steps for a highly structured decision
  - Humans taken out of decision

Examples:

- Financial Services: DBS Quick Finance
- E-commerce: Lazada/Shopee's dynamic pricing algorithms
- Manufacturing: Smart factory systems at Jurong Innovation District

What is Business Intelligence?

- Business intelligence
  - Infrastructure for collecting, storing, analyzing data produced by business
  - Databases, data warehouses, data marts
- Business intelligence vendors
  - Create business intelligence and analytics purchased by firms

Analytical Tools - Relationships, Patterns, Trends

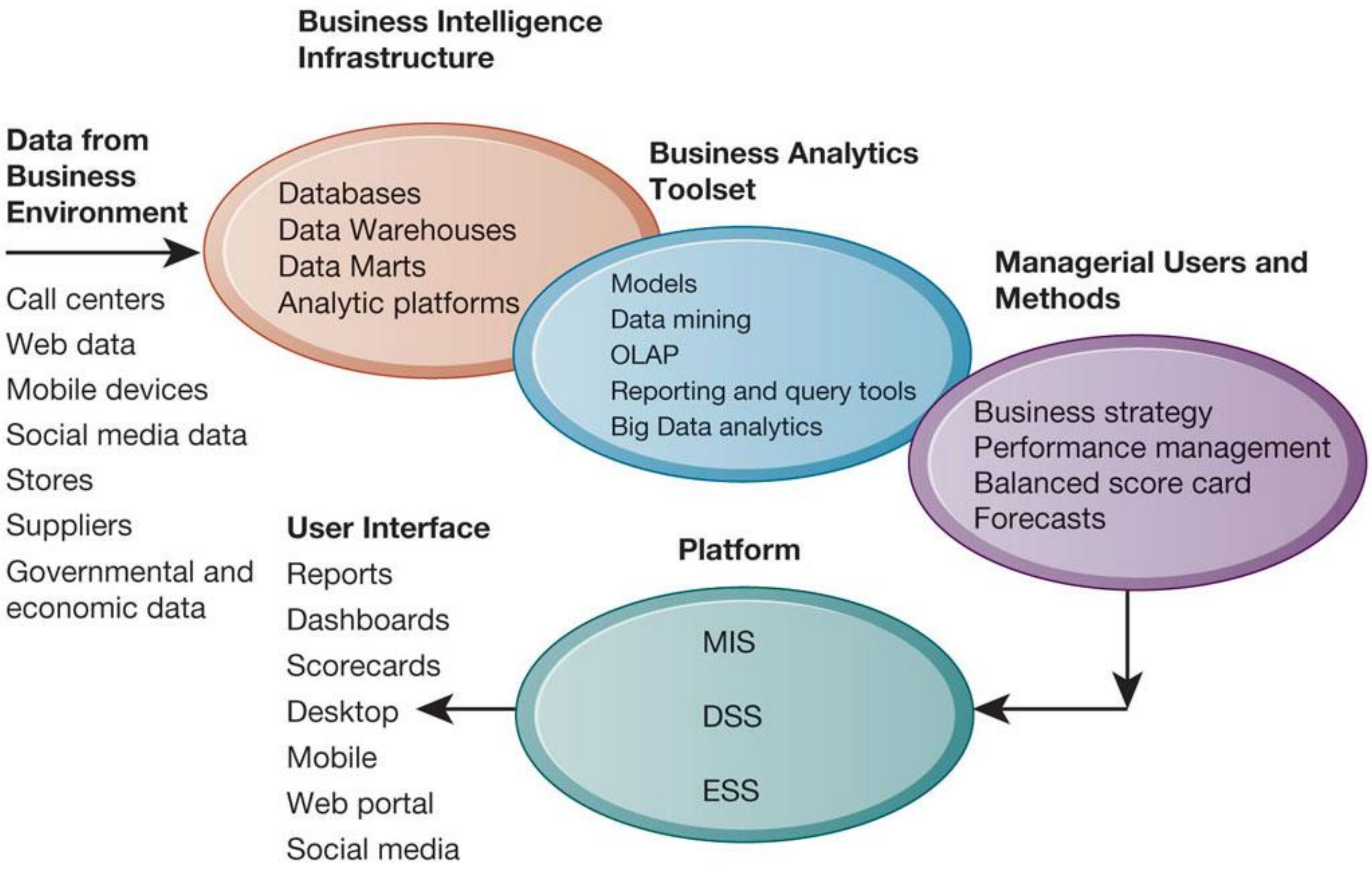
- Data warehouse
  - Stores current and historical data from many core operational transaction systems
  - Consolidates and standardizes information for use across enterprise, but data cannot be altered
- Data marts
  - Subset of data warehouse
  - Typically focus on single subject or line of business
- Business analytics
  - Tools and techniques for analyzing data
  - OLAP, statistics, models, data mining

The Business Intelligence Environment

Six elements in the business intelligence environment

1. Data from the business environment
2. Business intelligence infrastructure
3. Business analytics toolset
4. Managerial users and methods
5. Delivery platform—MIS, DSS, ESS
6. User interface
    - Data visualization tools

The Business Intelligence Environment



Business Intelligence and Analytics Capabilities

- Goal is to deliver accurate real-time information to decision makers
- Main analytic functionalities of BI systems
  - Production reports
  - Parameterized reports
  - Dashboards/scorecards
  - Ad hoc query/search/report creation
  - Drill down
  - Forecasts, scenarios, models

Business Intelligence and Analytics Capabilities

- Example

Business Functional Area	Production Reports
Sales	Forecast sales; sales team performance; cross-selling; sales cycle times
Service/call center	Customer satisfaction; service cost; resolution rates; churn rates
Marketing	Campaign effectiveness; loyalty and attrition; market basket analysis
Procurement and support	Direct and indirect spending; off-contract purchases; supplier performance
Supply chain	Backlog; fulfillment status; order cycle time; bill of materials analysis
Financials	General ledger; accounts receivable and payable; cash flow; profitability
Human resources	Employee productivity; compensation; workforce demographics; retention

Predictive Analytics

- Uses variety of data, techniques to predict future trends and behavior patterns
  - Statistical analysis
  - Data mining
  - Historical data
  - Assumptions
- Incorporated into numerous BI applications for sales, marketing, finance, fraud detection, health care
- Credit scoring
- Predicting responses to direct marketing campaigns

Predictive Analytics (cont.)

- Applications:
  - Customer churn prediction
  - Demand forecasting
  - Risk assessment
- Example: Bank predicting loan default probabilities based on customer behavior patterns

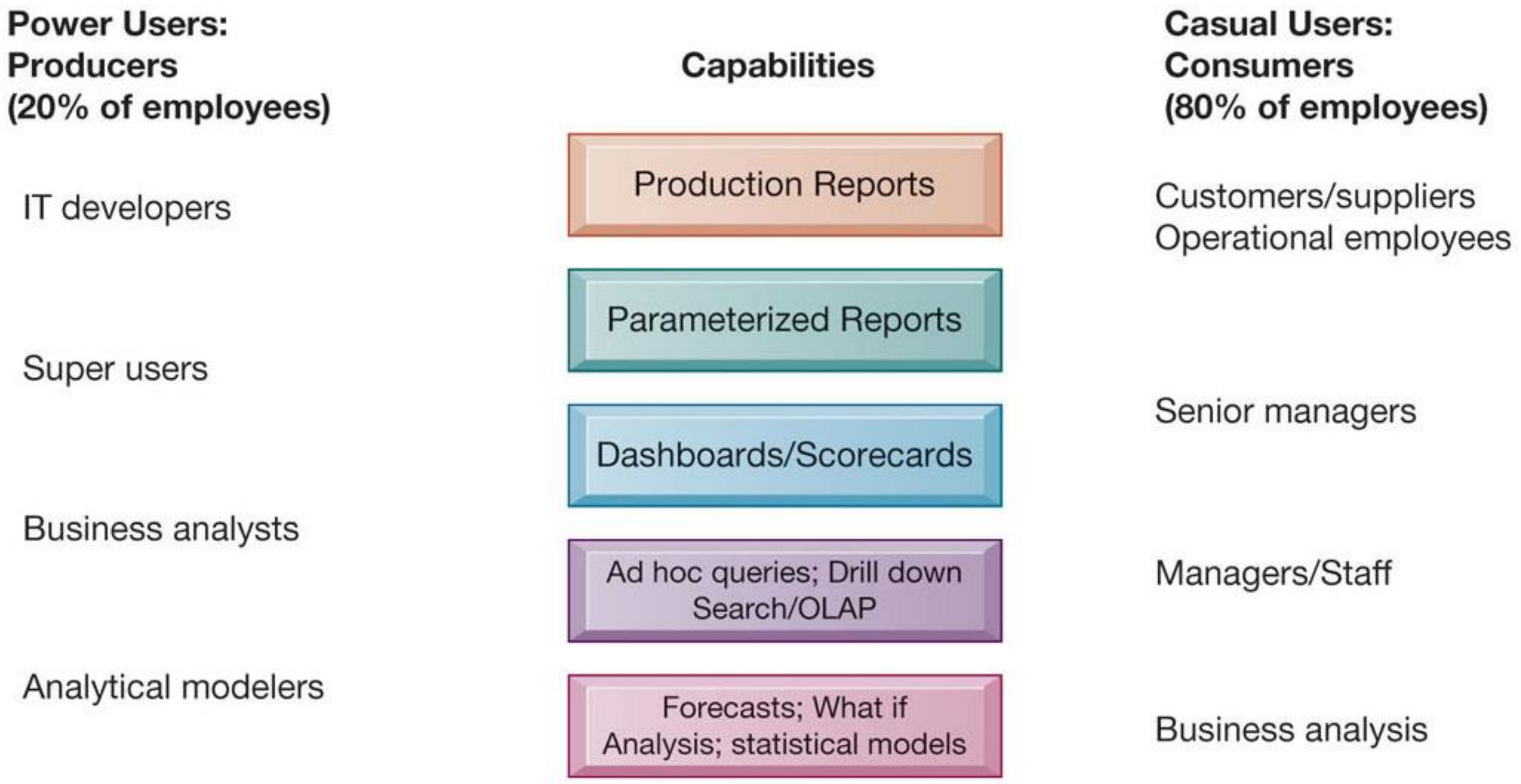
Operational Intelligence and Analytics

- Operational intelligence: Business activity monitoring
- Real-time analytics on business operations
- Collection and use of data generated by sensors
- Internet of Things (IoT) integration
    - Creating huge streams of data from web activities, sensors, and other monitoring devices
- Example: PSA Singapore's smart port operations:
    - Real-time tracking of container movements
    - Predictive maintenance of port equipment
    - Optimization of berth allocation and yard operations

Location Analytics and GIS

- Location analytics: Ability to gain business insight from the location (geographic) component of data
  - Mobile phones
  - Sensors, scanning devices
  - Map data
- Geographic information systems (GIS)
  - Ties location-related data to maps
- Example: Singapore Land Authority's OneMap integrating geospatial data for government and business applications

Business Intelligence Users



Support for Semi structured Decisions

- Decision-support systems
    - Support for semi structured decisions
- Use mathematical or analytical models
- Allow varied types of analysis
    - “What-if” analysis
    - Sensitivity analysis
    - Backward sensitivity analysis
    - Multidimensional analysis / OLAP
        - Example: pivot tables

Decision Support for Senior Management

- ESS: decision support for senior management
  - Help executives focus on important performance information
- Balanced scorecard method
  - Measures outcomes on four dimensions
    1. Financial
    2. Business process
    3. Customer
    4. Learning and growth
  - Key performance indicators (KPIs) measure each dimension

Decision Support for Senior Management (cont.)

- Business performance management (BPM)
  - Translates firm's strategies (e.g., differentiation, low-cost producer, scope of operation) into operational targets
  - KPIs developed to measure progress toward targets
- Data for ESS
  - Internal data from enterprise applications
  - External data such as financial market databases
  - Drill-down capabilities

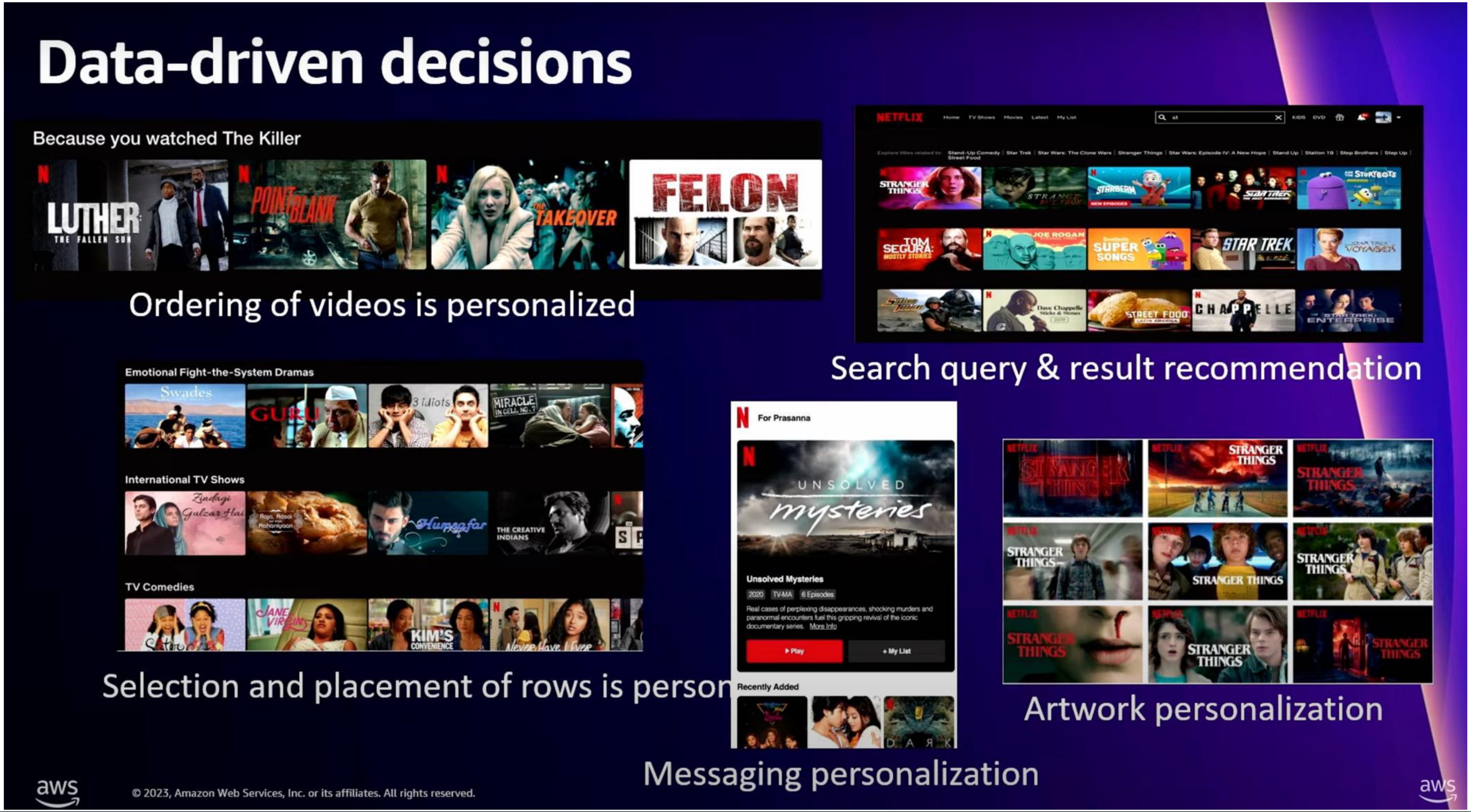
Business Performance Management

- Translating strategies into operational targets
- Monitoring KPIs related to strategic objectives
- Example: Airlines' performance dashboard tracking:
  - Financial metrics
  - Customer satisfaction
  - Operational efficiency
  - Employee productivity

Group Decision-Support Systems (GDSS)

- Interactive system to facilitate solution of unstructured problems by group
  - Facilitates collaborative decision making
- Specialized tools
  - Virtual collaboration rooms
  - Software to collect, rank, edit participant ideas and responses
- Promotes collaborative atmosphere, anonymity
- Cisco’s Collaboration Meeting Rooms Hybrid (CMR)
- Skype for Business

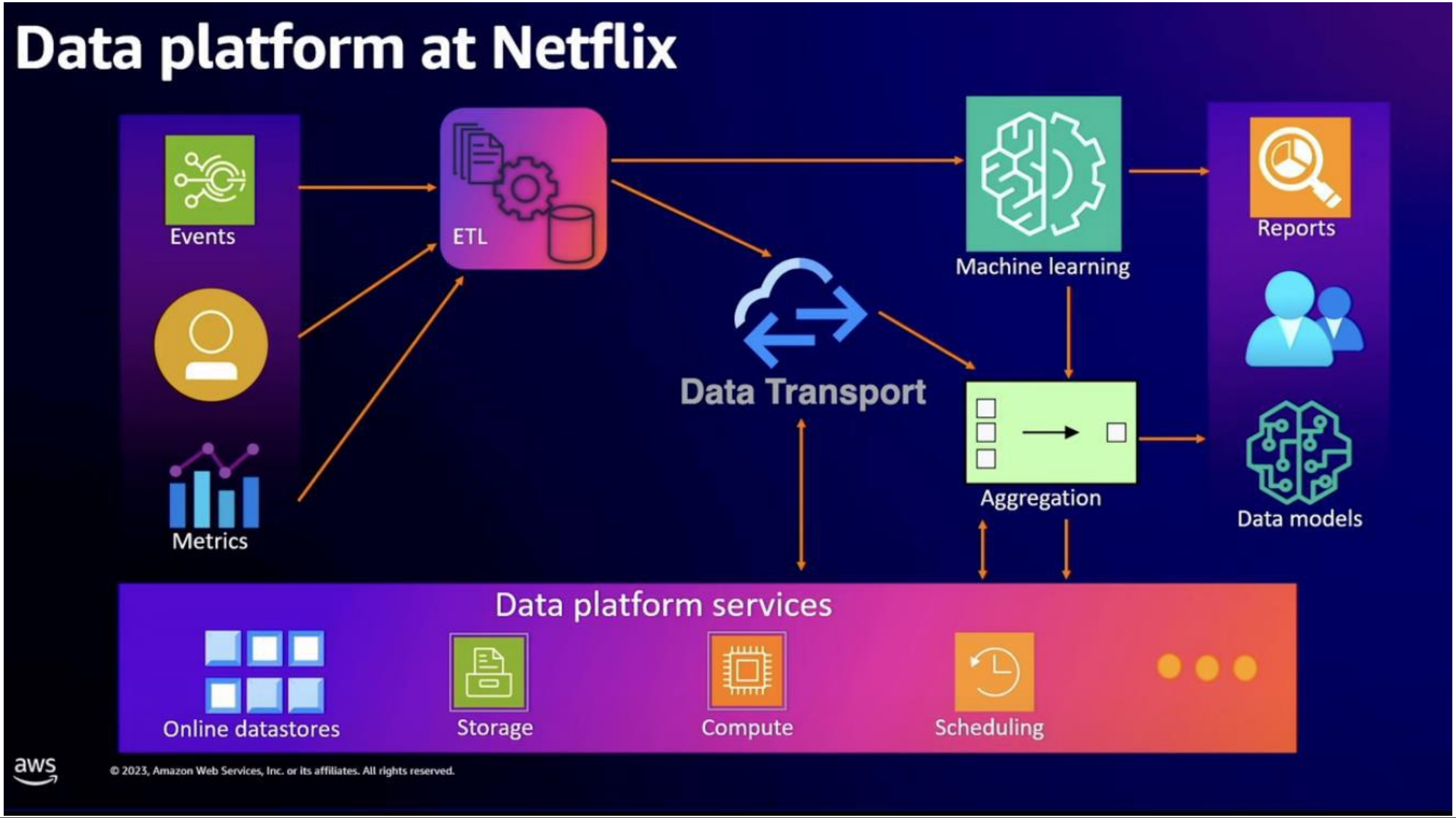
Example: Data Driven Decisions at Netflix



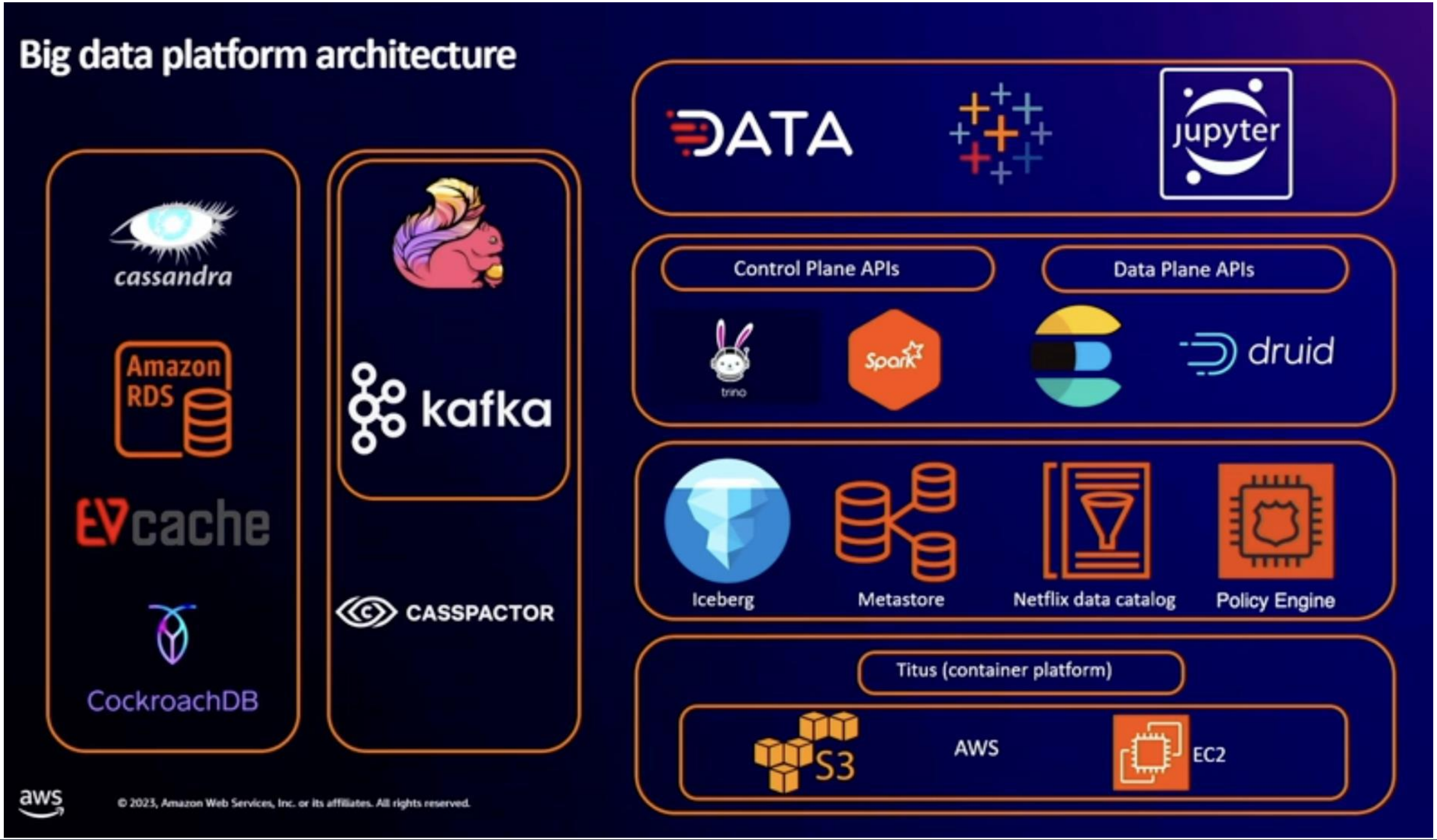
Example: Data Driven Decisions at Netflix

- Data-driven decision making at Netflix:
    - Content recommendation engine analyzing viewing patterns
    - Production decisions based on viewer preferences
    - A/B testing of user interface elements
    - Personalized marketing based on viewing history

Example: Data Driven Decisions at Netflix



Example: Data Driven Decisions at Netflix



Emerging Trends in Business Intelligence

- Artificial Intelligence and Machine Learning:
  - Automated insights generation
  - Natural language processing for data queries
  - Autonomous decision systems
- Augmented Analytics:
  - AI-assisted data preparation
  - Automated pattern detection
  - Natural language generation of reports

Challenges in Implementing BI

- Data quality and integration issues
- Skills gap in analytics talent
- Balancing automated and human decision making
- Ethical considerations in data usage
    - Example: GDPR, PDPA

Summary

- Business intelligence transforms raw data into actionable insights
- Different management levels require different types of decision support
- Modern BI encompasses traditional analytics, big data, and AI technologies
- Successful organizations balance automated and human decision making

NEXT

Building Information Systems